



How to SSH into your VM from another computer



Mando_elnino · [Follow](#)

4 min read · Mar 23, 2023



NAME: ABDULLAH SOHAIL SAP: 70147165

Welcome!

I'll be talking about how I was able to SSH into my Parrot VM from my Windows computer. Here will be a brief overview of the steps. I'll go more in depth as you read on.

1. Make sure you can ping the VM. If you can't, here's how you can configure that! [How to ping your VM that's in VirtualBox | by Mando_elnino | Mar, 2023 | Medium](#)
2. Install openssh-server if it's not already installed
3. Open Powershell from your Windows computer and try to ssh into the VM
4. Logout of the VM
5. STOP the openssh-server

First we will ping the VM from the Host OS. To do that run “ifconfig” on the virtual machine and look for the IP address that’s within the DNS range of your network.



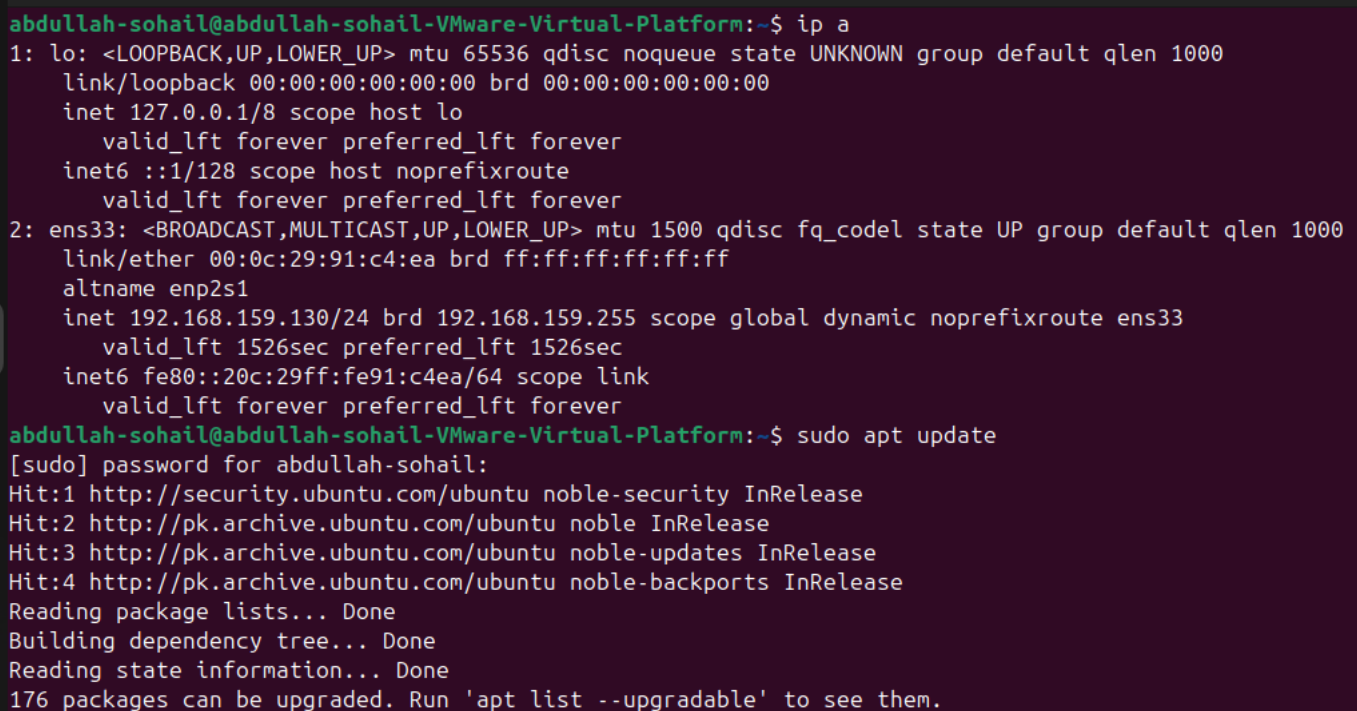
```
$ifconfig
enp0s8: flags=4163
    inet 192.168.1.100 netmask 255.255.255.0
    inet6 fe80::20c:29ff:fe91:c4ea/64 scope link
    inet6 2601::1000:1000:1000:1000/48 scope global
    inet6 2601::1000:1000:1000:1000/48 scope global
```

In my case the enp0s8 network interface has the IP address within the DNS range. This is what we will ping.

Now use tcpdump on the VM to detect the pings coming from your computer with this command

```
sudo tcpdump icmp
```

STEP 1:



```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:91:c4:ea brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.159.130/24 brd 192.168.159.255 scope global dynamic noprefixroute ens33
        valid_lft 1526sec preferred_lft 1526sec
    inet6 fe80::20c:29ff:fe91:c4ea/64 scope link
        valid_lft forever preferred_lft forever
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo apt update
[sudo] password for abdullah-sohail:
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://pk.archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://pk.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://pk.archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
176 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

and now open powershell on your computer and ping the IP that we checked for on the VM

```
ping <ip_address>
```

STEP 2:

```
PS C:\WINDOWS\system32> ping 192.168.159.130

Pinging 192.168.159.130 with 32 bytes of data:
Reply from 192.168.159.130: bytes=32 time=2ms TTL=64
Reply from 192.168.159.130: bytes=32 time<1ms TTL=64
Reply from 192.168.159.130: bytes=32 time<1ms TTL=64
Reply from 192.168.159.130: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.159.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms
PS C:\WINDOWS\system32>
```

```
$sudo tcpdump icmp
[sudo] password for wow:
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s8, link-type EN10MB (Ethernet), snapshot length 262144 bytes
19:12:24.150328 IP DESKTOP-RJC1V6M.lan > parrot.lan: ICMP echo request, id 1, seq 5, length 40
19:12:24.150421 IP parrot.lan > DESKTOP-RJC1V6M.lan: ICMP echo reply, id 1, seq 5, length 40
19:12:25.152067 IP DESKTOP-RJC1V6M.lan > parrot.lan: ICMP echo request, id 1, seq 6, length 40
19:12:25.152115 IP parrot.lan > DESKTOP-RJC1V6M.lan: ICMP echo reply, id 1, seq 6, length 40
19:12:26.158671 IP DESKTOP-RJC1V6M.lan > parrot.lan: ICMP echo request, id 1, seq 7, length 40
19:12:26.158711 IP parrot.lan > DESKTOP-RJC1V6M.lan: ICMP echo reply, id 1, seq 7, length 40
19:12:27.164564 IP DESKTOP-RJC1V6M.lan > parrot.lan: ICMP echo request, id 1, seq 8, length 40
19:12:27.164592 IP parrot.lan > DESKTOP-RJC1V6M.lan: ICMP echo reply, id 1, seq 8, length 40
```

We got a hit! Now that we verified they're on the same network we can move on.

2. Install OpenSSH server with this command from the VM

```
sudo apt-get install openssh-server
```

STEP 3:

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo apt install openssh-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ncurses-term openssh-client openssh-sftp-server ssh-import-id
Suggested packages:
  keychain libpam-ssh monkeysphere ssh-askpass molly-guard
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
The following packages will be upgraded:
  openssh-client
1 upgraded, 4 newly installed, 0 to remove and 175 not upgraded.
Need to get 1,738 kB of archives.
After this operation, 6,743 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://pk.archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-client amd64 1:9.6p1-3ubuntu13.14 [906 kB]
Get:2 http://pk.archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-sftp-server amd64 1:9.6p1-3ubuntu13.14 [37.3 kB]
Get:3 http://pk.archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-server amd64 1:9.6p1-3ubuntu13.14 [510 kB]
Get:4 http://pk.archive.ubuntu.com/ubuntu noble/main amd64 ncurses-term all 6.4+20240113-1ubuntu2 [275 kB]
Get:5 http://pk.archive.ubuntu.com/ubuntu noble-updates/main amd64 ssh-import-id all 5.11-0ubuntu2.24.04.1 [10.1 kB]
Fetched 1,738 kB in 5s (320 kB/s)
Preconfiguring packages ...
(Reading database ... 151601 files and directories currently installed.)
Preparing to unpack .../openssh-client_1%3a9.6p1-3ubuntu13.14_amd64.deb ...
Unpacking openssh-client (1:9.6p1-3ubuntu13.14) over (1:9.6p1-3ubuntu13.13) ...
Selecting previously unselected package openssh-sftp-server.
Preparing to unpack .../openssh-sftp-server_1%3a9.6p1-3ubuntu13.14_amd64.deb ...
Unpacking openssh-sftp-server (1:9.6p1-3ubuntu13.14) ...
Selecting previously unselected package openssh-server.
Preparing to unpack .../openssh-server_1%3a9.6p1-3ubuntu13.14_amd64.deb ...
Unpacking openssh-server (1:9.6p1-3ubuntu13.14) ...
Selecting previously unselected package ncurses-term.
Preparing to unpack .../ncurses-term_6.4+20240113-1ubuntu2_all.deb ...
Unpacking ncurses-term (6.4+20240113-1ubuntu2) ...
```

```
Get:3 http://pk.archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-server amd64 1:9.6p1-3ubuntu13.14 [510 kB]
Get:4 http://pk.archive.ubuntu.com/ubuntu noble/main amd64 ncurses-term all 6.4+20240113-1ubuntu2 [275 kB]
Get:5 http://pk.archive.ubuntu.com/ubuntu noble-updates/main amd64 ssh-import-id all 5.11-0ubuntu2.24.04.1 [10.1 kB]
Fetched 1,738 kB in 5s (320 kB/s)
Preconfiguring packages ...
(Reading database ... 151601 files and directories currently installed.)
Preparing to unpack .../openssh-client_1%3a9.6p1-3ubuntu13.14_amd64.deb ...
Unpacking openssh-client (1:9.6p1-3ubuntu13.14) over (1:9.6p1-3ubuntu13.13) ...
Selecting previously unselected package openssh-sftp-server.
Preparing to unpack .../openssh-sftp-server_1%3a9.6p1-3ubuntu13.14_amd64.deb ...
Unpacking openssh-sftp-server (1:9.6p1-3ubuntu13.14) ...
Selecting previously unselected package openssh-server.
Preparing to unpack .../openssh-server_1%3a9.6p1-3ubuntu13.14_amd64.deb ...
Unpacking openssh-server (1:9.6p1-3ubuntu13.14) ...
Selecting previously unselected package ncurses-term.
Preparing to unpack .../ncurses-term_6.4+20240113-1ubuntu2_all.deb ...
Unpacking ncurses-term (6.4+20240113-1ubuntu2) ...
Selecting previously unselected package ssh-import-id.
Preparing to unpack .../ssh-import-id_5.11-0ubuntu2.24.04.1_all.deb ...
Unpacking ssh-import-id (5.11-0ubuntu2.24.04.1) ...
Setting up openssh-client (1:9.6p1-3ubuntu13.14) ...
Setting up ssh-import-id (5.11-0ubuntu2.24.04.1) ...
Setting up ncurses-term (6.4+20240113-1ubuntu2) ...
Setting up openssh-sftp-server (1:9.6p1-3ubuntu13.14) ...
Setting up openssh-server (1:9.6p1-3ubuntu13.14) ...

Creating config file /etc/ssh/sshd_config with new version
Created symlink /etc/systemd/system/sockets.target.wants/ssh.socket → /usr/lib/systemd/system/ssh.socket.
Created symlink /etc/systemd/system/ssh.service.requires/ssh.socket → /usr/lib/systemd/system/ssh.socket.
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for ufw (0.36.2-6) ...
```

Check if ssh is running on the VM with this command

```
sudo systemctl status ssh
```

```
$ sudo systemctl status ssh
○ ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: inactive (dead)
     Docs: man:sshd(8)
           man:sshd_config(5)
```

STEP NO 4:

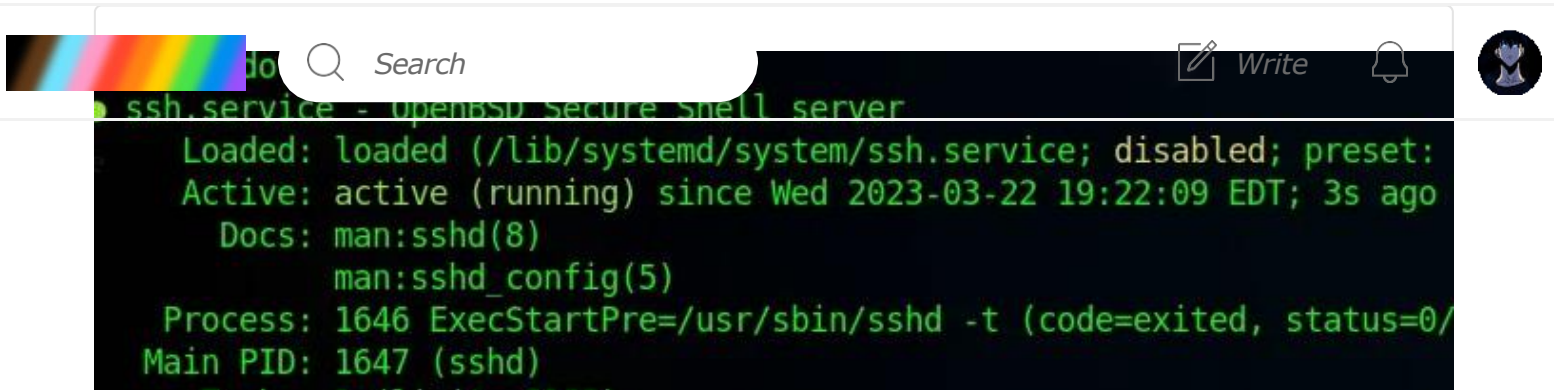
```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo systemctl status ssh
○ ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: inactive (dead)
TriggeredBy: ● ssh.socket
   Docs: man:sshd(8)
        man:sshd_config(5)
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo systemctl start ssh
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Tue 2026-02-17 11:59:36 PKT; 10s ago
TriggeredBy: ● ssh.socket
   Docs: man:sshd(8)
        man:sshd_config(5)
  Process: 5806 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
 Main PID: 5807 (sshd)
    Tasks: 1 (limit: 9378)
   Memory: 1.2M (peak: 1.4M)
      CPU: 25ms
   CGroup: /system.slice/ssh.service
           └─5807 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Feb 17 11:59:36 abdullah-sohail-VMware-Virtual-Platform systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Feb 17 11:59:36 abdullah-sohail-VMware-Virtual-Platform sshd[5807]: Server listening on 0.0.0.0 port 22.
Feb 17 11:59:36 abdullah-sohail-VMware-Virtual-Platform sshd[5807]: Server listening on :: port 22.
Feb 17 11:59:36 abdullah-sohail-VMware-Virtual-Platform systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
```

We see that it's inactive. So we will temporarily start it and stop it once we're finished. Check the status

To enable it run this command

[Open in app](#)



It's active! Now let's try and ssh into it.

3. Now open Powershell and attempt to ssh into the VM with this command.
The IP address will be the address that we pinged in the first step.

```
ssh machine_username@ip_address
```

STEP 5:

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ ssh abdullah-sohail@192.168.159.130
The authenticity of host '192.168.159.130 (192.168.159.130)' can't be established.
ED25519 key fingerprint is SHA256:Yn5qIoNsGCN8TIVaQRatNYG/xAIIMVabQPULHGMEOKA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.159.130' (ED25519) to the list of known hosts.
abdullah-sohail@192.168.159.130's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.17.0-14-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

168 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$
```

```
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])?
```

You'll be asked if you want to continue, put "yes"

```
password:
```

Enter the password. The command line will not show any letters that you're typing in an effort to increase the security. Don't be alarmed if you're typing and nothing is there, the password will be accepted if it's correct.

```
Linux parrot 6.0.0-12parrot1-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.0.12-1parrot1 (2023-01-12) x86_64

[ ASCII Art ]

The programs included with the Parrot GNU/Linux are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Parrot GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
[wow@parrot]~$
```

BOOM!

We're in the VM! We can now run commands as if we were in the VM, except we're on another computer! SUPER COOL

4. Logout of Parrot by typing exit in the windows machine and you'll be prompted back to the powershell command line.

```
$exit
logout
```

5. Now we will go back to the VM to stop the sshserver using this command

```
sudo systemctl stop ssh
```

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo systemctl stop ssh
[sudo] password for abdullah-sohail:
Warning: The unit file, source configuration file or drop-ins of ssh.service changed on disk. Run 'systemctl daemon-reload' to reload units.
Stopping 'ssh.service', but its triggering units are still active:
ssh.socket
```

now check the status running this command

```
sudo systemctl status ssh
```

You will see that not only did we stop the service(showing it's inactive), it recorded a history of our logon from the other device

```
$ sudo systemctl status ssh
○ ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; disabled; pres
   Active: inactive (dead)
     Docs: man:sshd(8)
           man:sshd_config(5)

Mar 22 19:22:09 parrot systemd[1]: Starting ssh.service - OpenBSD Se
Mar 22 19:22:09 parrot sshd[1647]: Server listening on 0.0.0.0 port
Mar 22 19:22:09 parrot sshd[1647]: Server listening on :: port 22.
Mar 22 19:22:09 parrot systemd[1]: Started ssh.service - OpenBSD Sec
Mar 22 19:27:58 parrot sshd[1662]: Accepted password for wow from 19
Mar 22 19:27:58 parrot sshd[1662]: pam_unix(sshd:session): session o
Mar 22 19:36:45 parrot systemd[1]: Stopping ssh.service - OpenBSD Se
Mar 22 19:36:45 parrot systemd[1]: ssh.service: Deactivated successf
Mar 22 19:36:45 parrot systemd[1]: Stopped ssh.service - OpenBSD Sec
lines 1-15/15 (END)
```

It's very important to stop the service because it is a VERY big security risk to leave it open. DO NOT LEAVE SSH OPEN PLEASE!

STEP 6:

```
Warning: The sshd config file is not writable by root. Stop the sshd
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ exit
logout
Connection to 192.168.159.131 closed.
```

Some benefits of being able to access another device by ssh-ing into it is that you can be anywhere in the world and connect to a remote device such as a work device. However one question is how will they connect if they're not on the same network interface? One way to ensure that is to have both of the devices connected to the same openvpn network. If you've ever noticed in hack the box or tryhackme you must connect to an openvpn in order to access the machine. The same works here. If someone were in England but they had to connect to their work device in New York, they can have someone connect to the VPN on their workstation in New York, the person in England would connect to the same VPN(using the same vpn file. Not just using the same brand of VPN) on their computer, and then they're on the same network interface! This idea of being able to connect to a remote device is awesome and I had a lot of fun doing it in my own home. Just remember you have to stop the ssh service!! Once the work is done on the remote system there's no need to have the service running anymore.

I appreciate any comments and as always have fun :)

Ssh

Virtualization

Parrotos

Powershell

Virtualbox



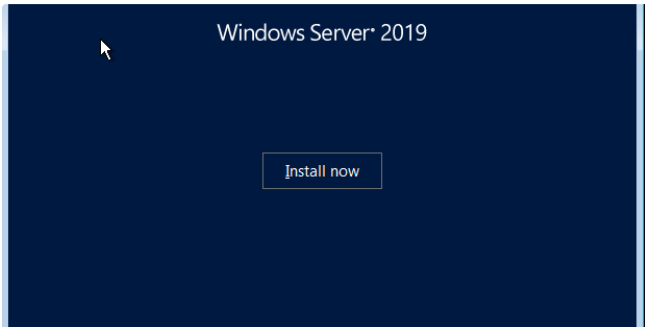
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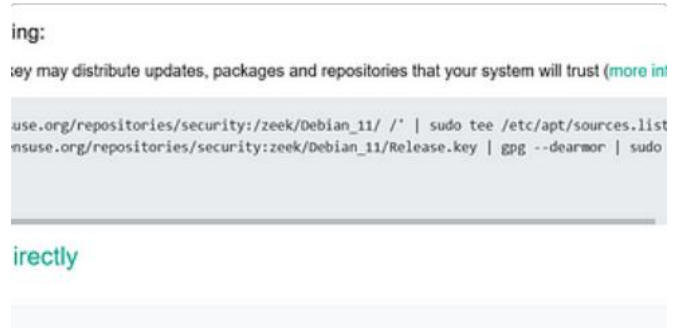
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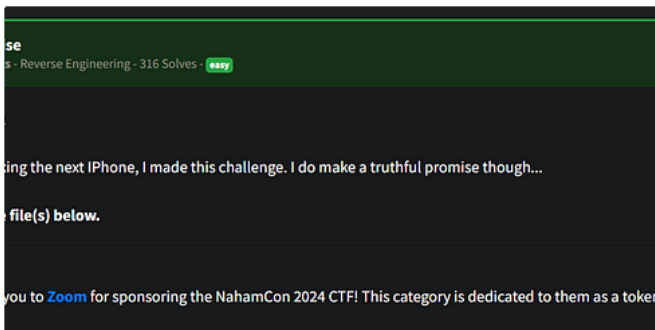
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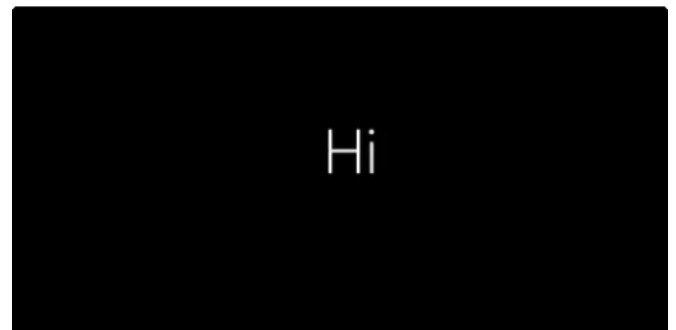
 



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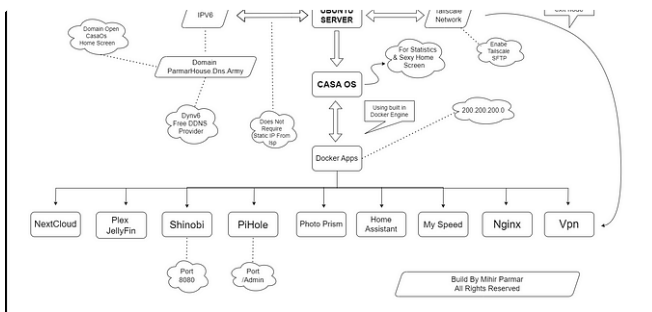
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